



Methods in Calculus · Mark Scheme

Core Pure 2 Chapter 3: Methods in Calculus. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Improper integrals

- 1 $\lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_1^t = \lim_{t \rightarrow \infty} \left(1 - \frac{1}{t} \right)$ M1; = 1 A1.
- 2 $x^{-\frac{1}{3}}$ is undefined at $x = 0$ B1; $\lim_{t \rightarrow 0} \left[\frac{3}{2}x^{\frac{2}{3}} \right]_t^8 = \lim_{t \rightarrow 0} \left(6 - \frac{3}{2}t^{\frac{2}{3}} \right)$ M1; = 6 A1.
- 3 $\int_1^t x^{-\frac{1}{2}} dx = [2\sqrt{x}]_1^t$ M1; $= 2\sqrt{t} - 2$ A1; $2\sqrt{t} \rightarrow \infty$ as $t \rightarrow \infty$, so the integral diverges A1.
- 4 Parts: $\int xe^{-x} dx = -xe^{-x} - e^{-x}$ M1 A1; $[-xe^{-x} - e^{-x}]_0^t = 1 - te^{-t} - e^{-t}$ M1; $\rightarrow 1$ as $t \rightarrow \infty$ A1.
- 5(a) $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$ B1; $\left[\ln \frac{x}{x+1} \right]_1^t = \ln \frac{t}{t+1} - \ln \frac{1}{2}$ M1; $\frac{t}{t+1} \rightarrow 1$ so $\ln \frac{t}{t+1} \rightarrow 0$ as $t \rightarrow \infty$ M1; integral = $-\ln \frac{1}{2} = \ln 2$ A1.
- 5(b) Parts with $u = \ln x$, $v' = 1$: $\int \ln x dx = x \ln x - x$ M1 A1; $[x \ln x - x]_t^1 = -1 - t \ln t + t$ M1; $\rightarrow -1$ as $t \rightarrow 0$ A1.

Section B – Mean value of a function

- 6 $\frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3} \times 9$ M1; = 3 A1.
- 7 $\frac{1}{\pi - 0} \int_0^\pi \sin x dx$ M1; $[-\cos x]_0^\pi = 2$ A1; mean value $\frac{2}{\pi}$ A1.
- 8 $\frac{1}{1} \int_0^1 \frac{1}{1+x^2} dx = [\arctan x]_0^1$ M1 A1; = $\frac{\pi}{4}$ A1.
- 9 Parts: $\int xe^x dx = xe^x - e^x$ M1 A1; $[(x-1)e^x]_0^2 = e^2 + 1$ A1; mean value $\frac{1}{2}(e^2 + 1)$ A1.
- 10(a) $\tan y = x \Rightarrow \sec^2 y \frac{dy}{dx} = 1$ M1; $\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$ A1.
- 10(b) $\int_0^2 \frac{4}{4+x^2} dx = \left[2 \arctan \frac{x}{2} \right]_0^2$ M1 A1; = $\frac{\pi}{2}$, so mean value = $\frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$ A1.
- 10(c) (i) $\frac{\pi}{4} + 3$ B1; (ii) $-\frac{\pi}{2}$ B1.

Section C – Arc length

- 11 $\frac{dy}{dx} = \sinh x$ B1; $1 + \sinh^2 x = \cosh^2 x$ B1.
- 12 $s = \int_0^{\ln 2} \cosh x dx = [\sinh x]_0^{\ln 2}$ M1; $\sinh(\ln 2) = \frac{1}{2} \left(2 - \frac{1}{2} \right)$ M1; = $\frac{3}{4}$ A1.
- 13 $\frac{dy}{dx} = x^{\frac{1}{2}}$, so $s = \int_0^3 \sqrt{1+x} dx$ M1; $= \left[\frac{2}{3} (1+x)^{\frac{3}{2}} \right]_0^3$ A1; = $\frac{2}{3} (8-1) = \frac{14}{3}$ A1.
- 14 $\dot{x} = 2t$, $\dot{y} = 2t^2$ B1; $\sqrt{4t^2 + 4t^4} = 2t\sqrt{1+t^2}$ (as $t \geq 0$) M1; $s = \left[\frac{2}{3} (1+t^2)^{\frac{3}{2}} \right]_0^{2\sqrt{2}}$ A1; = $\frac{2}{3} (27-1) = \frac{52}{3}$ A1.
- 15(a) $\dot{x} = -3 \cos^2 t \sin t$, $\dot{y} = 3 \sin^2 t \cos t$ B1; $\dot{x}^2 + \dot{y}^2 = 9 \sin^2 t \cos^2 t (\cos^2 t + \sin^2 t)$ M1; = $9 \sin^2 t \cos^2 t$ A1.
- 15(b) $s = \int_0^{\frac{\pi}{2}} 3 \sin t \cos t dt$ (positive root, since $\sin t \cos t \geq 0$ in the first quadrant) M1 A1; = $\left[\frac{3}{2} \sin^2 t \right]_0^{\frac{\pi}{2}}$ M1; = $\frac{3}{2}$ A1.



15(c) By symmetry, $4 \times \frac{3}{2} = 6$ A1.

Section D – Surface area of revolution

$$16 \quad S = \int_0^2 2\pi x \sqrt{2} dx \text{ M1}; = 2\sqrt{2}\pi \times 2 = 4\sqrt{2}\pi \text{ A1}.$$

$$17 \quad \frac{dy}{dx} = x^{-\frac{1}{2}}, \text{ so } y\sqrt{1+\frac{1}{x}} = 2\sqrt{x+1} \text{ M1}; S = 4\pi \int_0^3 \sqrt{x+1} dx = 4\pi \left[\frac{2}{3}(x+1)^{\frac{3}{2}} \right]_0^3 \text{ A1}; = \frac{56\pi}{3} \text{ A1}.$$

$$18 \quad \frac{dy}{dx} = \frac{-x}{\sqrt{r^2-x^2}} \text{ B1}; y\sqrt{1+\frac{x^2}{r^2-x^2}} = \sqrt{r^2-x^2} \cdot \frac{r}{\sqrt{r^2-x^2}} = r \text{ M1}; S = \int_{-r}^r 2\pi r dx = 4\pi r^2 \text{ A1}.$$

$$19 \quad S = 2\pi \int_0^{\ln 2} \cosh x \cdot \cosh x dx \text{ M1}; = \pi \int_0^{\ln 2} (1 + \cosh 2x) dx = \pi \left[x + \frac{1}{2} \sinh 2x \right]_0^{\ln 2} \text{ M1 A1}; \sinh(2 \ln 2) = \frac{1}{2} \left(4 - \frac{1}{4} \right) = \frac{15}{8},$$

so $S = \pi \left(\ln 2 + \frac{15}{16} \right) \text{ A1}.$

$$20(a) \quad \dot{x} = 6t, \dot{y} = 3 - 3t^2 \text{ B1}; 36t^2 + 9 - 18t^2 + 9t^4 = 9(1 + 2t^2 + t^4) \text{ M1}; = 9(1 + t^2)^2 \text{ A1}.$$

$$20(b) \quad s = \int_0^{\sqrt{3}} 3(1+t^2) dt = [3t + t^3]_0^{\sqrt{3}} \text{ M1}; = 3\sqrt{3} + 3\sqrt{3} = 6\sqrt{3} \text{ A1}.$$

$$20(c) \quad S = 2\pi \int_0^{\sqrt{3}} (3t - t^3) \cdot 3(1+t^2) dt \text{ M1};$$

$$= 6\pi \int_0^{\sqrt{3}} (3t + 2t^3 - t^5) dt \text{ A1}; = 6\pi \left[\frac{3}{2}t^2 + \frac{1}{2}t^4 - \frac{1}{6}t^6 \right]_0^{\sqrt{3}} \text{ M1}; = 6\pi \left(\frac{9}{2} + \frac{9}{2} - \frac{9}{2} \right) = 27\pi \text{ A1}.$$

Roar-level challenge

$$21 \quad x^2 + 2x + 5 = (x+1)^2 + 4 \text{ B1}; \int \frac{1}{(x+1)^2 + 4} dx = \frac{1}{2} \arctan \frac{x+1}{2} \text{ M1 A1}; \text{ split, e.g. at } x = -1: \lim_{t \rightarrow \infty} \frac{1}{2} \arctan \frac{t+1}{2} -$$

$$\lim_{s \rightarrow -\infty} \frac{1}{2} \arctan \frac{s+1}{2}, \text{ both limits finite, so it converges M1}; = \frac{1}{2} \left(\frac{\pi}{2} \right) - \frac{1}{2} \left(-\frac{\pi}{2} \right) = \frac{\pi}{2} \text{ A1}.$$